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Not Just for Tough Times: The Efficacy and Mechanisms of Positive Goal Reappraisal in Negative, Neutral, and Positive Contexts

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Reappraisal is a common emotion regulation strategy that involves adjusting how a situation is appraised. While much is known about its use to reduce negative affect in negative situations, less is known about its use across negative, neutral, and positive contexts to increase positive affect (i.e., positive goal reappraisal). To fill this gap, we investigated the efficacy and mechanisms of positive goal reappraisal across three valence categories in two complementary studies. In Study 1, 158 participants rated their subjective affect and reported how they appraised depicted situations on key appraisal dimensions with and without using reappraisal. In Study 2, 70 participants completed the same task, while their electromyographic and electroencephalographic responses were recorded. We found that reappraisal was effective across all valence categories, as it increased subjective positive affect and decreased subjective negative affect in response to negative, neutral, and positive pictures. Reappraisal also increased *zygomaticus major* reactivity for neutral and positive pictures and decreased *corrugator supercilii* reactivity for negative and neutral pictures. Regarding cognitive mechanisms, we found that the effects of reappraisal were related to appraisal shifts, particularly changes in congruence and relevance. A broader range of appraisal shifts were involved in neutral and positive contexts than in negative ones, suggesting that reappraisal mechanisms may be context-dependent. Finally, reappraisal amplified the late positive potential across all picture types within a relatively early time window (263–1,013 ms), indicating sustained attentional engagement. We conclude that positive goal reappraisal may be effective irrespective of stimulus valence by producing appraisal shifts.

Keywords: reappraisal, appraisal dimensions, emotion regulation, facial electromyography, late positive potential

Supplemental materials: https://doi.org/10.1037/emo0001613.supp

One widely used emotion regulation strategy is reappraisal, which involves changing how one thinks about a potentially emotion-eliciting situation in order to change the emotional response (McRae, 2016; A. Uusberg et al., 2019). Most experimental studies have focused on reappraisal aimed at reducing negative affect in negative situations. However, emotion regulation can promote emotional well-being across a broader range of situations (e.g., Aldao & Tull, 2015; Petrova & Gross, 2023). In this article, we extend prior work by investigating the efficacy and cognitive mechanisms of reappraisal aimed at increasing positive affect in negative, neutral, and positive contexts.

Efficacy of Reappraisal in Different Valence Contexts

Prototypically seen as a way to deal with negative emotions, reappraisal is in fact applicable across a wide range of situations with different initial emotional valence. For instance, imagine playing a game like tennis. You might feel frustrated when you

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miss a shot, happy when you win a point, and neutral when preparing for the game. In each of these situations, you can use reappraisal to improve your affect. You might reappraise missing a shot by viewing it as a valuable learning opportunity. You could put an even more positive spin on winning a point by interpreting it as a sign that your training is paying off. In the neutral situation of preparing for the game, you might remind yourself of the opportunities the upcoming game holds for your career. Even though the initial valence of these situations differs, reappraisal has the potential to improve affect in each of them.

In different valence contexts, reappraisal can serve different emotion regulation goals—desired emotional states that individuals are motivated to achieve (McRae et al., 2012; Tamir, 2016). In negative situations, individuals usually aim to reduce the intensity or duration of their negative emotions (Gross, 2015). This goal of decreasing negative affect is unlikely to arise in positive or neutral situations where negative affect is minimal or absent. However, emotion regulation can also enhance positive affect—a goal that may be relevant across all contexts. In positive situations, people may wish to intensify or prolong the positive affect they already experience. In neutral situations, people may aim to generate positive affect where none exists. Even in negative situations, they not only seek to reduce negative affect but may also try to replace or complement it with positive affect, which can be more effective for modulating overall affect than merely decreasing negative affect (e.g., Kreibig et al., 2023). Reappraisal aimed at increasing positive affect (i.e., positive goal reappraisal) can thus be an important resource for promoting resilience and psychological well-being across different valence contexts (Lyubomirsky et al., 2005; McRae & Mauss, 2016).

While positive goal reappraisal has been found effective in negative situations, little is known about how it functions in neutral and positive contexts. Previous experimental research has shown that positive goal reappraisal can increase positive affect in response to both negative and positive stimuli and decrease negative affect in response to negative stimuli (e.g., Giuliani et al., 2008; Kreibig et al., 2023; Li et al., 2018; McRae et al., 2012; Samson et al., 2014; vanOyen Witvliet et al., 2010; Vlasenko et al., 2024; Waugh et al., 2016). However, to our knowledge, the effects of positive goal reappraisal have not been studied in response to neutral stimuli. Yet, increasing positive affect in neutral contexts is an important research target, given how often people experience and aim to improve neutral states (Gasper et al., 2019; Tamir, 2009).

Moreover, no studies have directly compared the efficacy of positive goal reappraisal across negative, neutral, and positive contexts. It is thus unclear which valence context is most conducive to positive goal reappraisal. It might be most effective in positive contexts, which afford more positive interpretations than negative or neutral ones. Alternatively, it could be most effective in negative contexts, where there is more room to improve how one feels. Finally, it may be most effective in neutral contexts, where its implementation is unhindered by the ceiling effect that may occur in positive contexts or by the conflicting initial affective state present in negative contexts. To address these open questions, the first aim of the present research was to directly compare the effects of positive goal reappraisal in response to negative, neutral, and positive stimuli, using both subjective and psychophysiological measures of affect.

Cognitive Mechanisms of Reappraisal in Different Valence Contexts

Reappraisal can be implemented using various cognitive actions (Garnefski & Kraaij, 2007; McRae et al., 2012; Vishkin et al., 2020). Building on appraisal theories (Moors et al., 2013; Roseman, 2013; Scherer, 1999; Smith & Lazarus, 1993), we have proposed (A. Uusberg et al., 2019) and found empirical support (A. Uusberg et al., 2023) for the idea that reappraisal alters emotions through shifts along appraisal dimensions that represent abstract evaluations of a situation. Although authors study slightly different sets of appraisal dimensions, five—relevance, congruence, accountability, certainty, and controllability—appear in most appraisal theories and are therefore considered necessary by many authors (Moors et al., 2013). Shifts along these dimensions can be used to describe and compare the mechanisms underlying different instances of reappraisal.

Depending on the situation, reappraisal may involve shifting different appraisal dimensions to improve affect. Returning to the example of playing tennis, you might remind yourself of the opportunities the game holds for your career, thereby increasing the appraised *relevance* of the situation (i.e., the importance of the situation for salient motives). When you reinterpret missing a shot as a learning opportunity, you shift the *congruence* appraisal (i.e., the extent to which the situation helps or hinders pursuing salient motives) by aligning the situation more closely with an important goal. Reflecting on how your training is paying off after winning a point increases your sense of *accountability* (i.e., the locus of responsibility for the situation). Alternatively, you might modify the *certainty* appraisal (i.e., the clarity of the situation and its implications) by considering why you might win the game. Finally, you could increase the *controllability* appraisal (i.e., the likelihood of changing the situation or adapting to its effects) by reminding yourself of your ability to make a strong comeback after losing a set.

It is unclear to what extent the efficacy of different appraisal shifts depends on the initial valence of a situation. For some appraisal dimensions, affect may improve through shifts in the same direction across all valence contexts. For instance, increased *congruence* is likely to improve affect in negative, neutral, and positive contexts, as individuals reassess how the situation aligns with their broader goals (Moors et al., 2013; Scherer, 1999). For other dimensions, however, affect may improve through shifts in opposite directions depending on the valence context. For example, in negative contexts, affect may improve when the appraised *relevance* of the situation is decreased, whereas in positive contexts, it may improve when *relevance* is increased. To explore these potentially context-dependent mechanisms, the second aim of the present research was to investigate which appraisal shifts are involved in positive goal reappraisal in response to negative, neutral, and positive stimuli.

Present Research

We investigated the efficacy of positive goal reappraisal in response to negative, neutral, and positive stimuli (Research Question 1), as well as the mechanistic role of appraisal shifts (Research Question 2) in two complementary studies. In online Study 1, 158 participants viewed negative, neutral, and positive pictures with and without using reappraisal to feel more positive. After each picture, participants rated their subjective positive and negative affect and described how they appraised the situation depicted in the picture. In a laboratory replication Study 2, 70 participants completed the same task while their facial electromyography (EMG) and electroencephalographic (EEG) signals were recorded. In both studies, participants imagined being personally present in situations depicted in perceptually balanced sets of standardized affective pictures, which yielded a range of relatively controlled affective states. For Research Question 1, we used linear mixed models (LMMs) to examine how positive goal reappraisal impacted ratings (Studies 1 and 2) and psychophysiological measures (Study 2) of positive and negative affect across negative, neutral, and positive stimuli. For Research Question 2, we used latent change score modeling (LCSM; Ferrer & McArdle, 2010; McArdle, 2009) to identify which appraisal shifts were involved in the effects of reappraisal across negative, neutral, and positive stimuli.

Transparency and Openness

Both studies were approved by the Research Ethics Committee of the University of Tartu, Estonia (369/T-7 and 379/T-1). We preregistered all measures, methods, sample size decisions, exclusion criteria, EEG and EMG preprocessing steps, and statistical analyses (Study 1 at https://osf.io/8vawm and Study 2 at https://osf.io/p3usv). Minor adjustments were made to the EEG and EMG preprocessing and statistical analyses as detailed below. All Supplemental Materials, along with preprocessed data and analysis code, are available at https://osf.io/3f96a/.

Study 1: Online Experiment

We conducted an online experiment to recruit a large and relatively heterogeneous sample of 158 healthy adults, allowing us to detect potentially small effects. Participants rated their subjective affect and reported how they appraised situations depicted in negative, neutral, and positive pictures, both with and without using reappraisal.

Method

Participants

Participants were volunteers recruited via the Prolific Academic survey platform in exchange for $15 in accordance with Prolific Academic's ethical payment principles. Only users with a minimum age of 18 years, who were fluent in English, had no ongoing mental health illness or condition, did not use medication to treat symptoms of depression, anxiety, or low-mood, and had an approval rate of ≥ 99% were invited to take part in the study. We recruited 165 participants to achieve 80% power to detect within-subject differences of $d = 0.22$. All participants met the attention check criterion; however, seven were excluded for failing to meet the comprehension and adherence criterion for the reappraisal task (see "Procedure and Design"). The final sample consisted of 158 participants (83 female, 75 male; age range 18 and 59 years, $M = 27.7$, $SD = 8.1$). Fifty-two percent of participants identified as White, 26% as Black, 2% as Asian, 12% as multiethnic, and the remaining 8% as some other ethnicity.

Stimuli

A total of 144 pictures (48 negative, 48 neutral, and 48 positive) were selected from the International Affective Picture System (IAPS; Lang et al., 2005), the Nencki Affective Picture System (NAPS; Marchewka et al., 2014), the Geneva Affective Picture Database (GAPED; Dan-Glauser & Scherer, 2011), and the Emotional Picture Set (EmoPicS; Wessa et al., 2010). These pictures were distributed across three parallel sets matched for normative valence and arousal ratings. The sets as well as valence categories were also matched for semantic content (e.g., portraits of a child) and physical properties (e.g., spatial frequency, luminance). See Supplemental Materials 1.1–1.3 for picture codes, descriptive statistics of the stimulus sets, and picture-level data collected in this study.

Procedure and Design

Study 1 consisted of an experimental task on the Pavlovia platform and a postexperiment questionnaire on the https://formr.org platform. Data were collected in March 2023.

After providing informed consent, participants completed the 2 (condition: VIEW, REAPPRAISE) by 3 (valence category: negative, neutral, positive) within-subject experiment. Participants viewed pictures in four blocks in a fixed order: VIEW, REAPPRAISE, VIEW, and REAPPRAISE. Block design was used to enhance adherence to task instructions and to reduce the cognitive load associated with completing self-report measures. One randomly selected parallel set of 16 negative, 16 neutral, and 16 positive pictures was presented during the VIEW condition, and another during the REAPPRAISE condition. Within each block, a random subset of eight negative, eight neutral, and eight positive pictures was presented in pseudorandomized order, ensuring that no more than two stimuli from the same valence category appeared consecutively. In total, each participant saw 96 pictures.

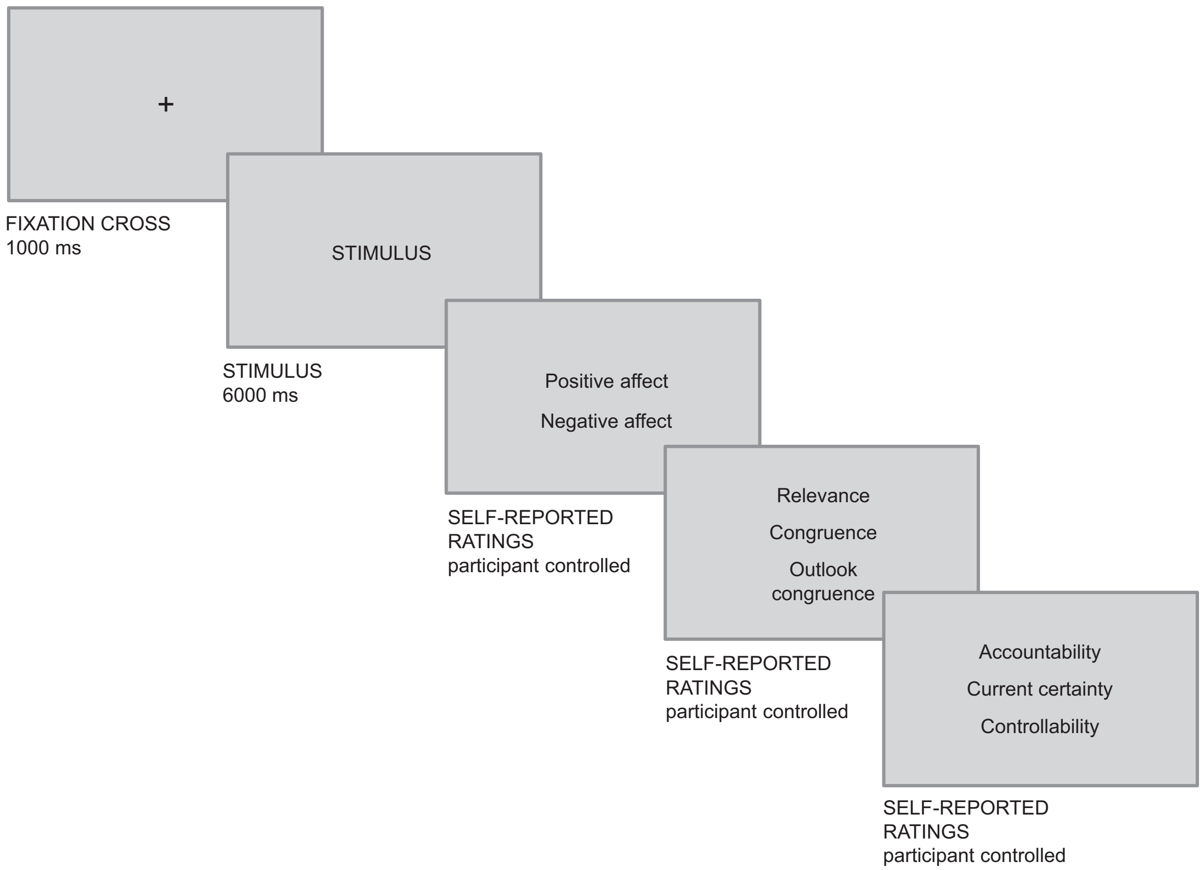
Each block began with written instructions (see Supplemental Materials 1.4). During the VIEW condition, participants were instructed to view the pictures while imagining that they were personally present, witnessing the depicted situation, and allowing any thoughts and feelings to arise as they naturally would, without trying to change them. During the REAPPRAISE condition, participants were instructed to imagine being personally present, witnessing the depicted situation, but this time to actively try to change how they think about the situation to feel more positive. Participants could reconsider what was going on in the situation, such as why the situation might be better than it seemed or how it could resolve or improve. Participants could also re-evaluate how much the situation mattered to them, reflecting on why this situation might be more or less important than they initially thought.

The first VIEW and REAPPRAISE blocks began with three practice trials using one negative, one neutral, and one positive picture that were not included in the experiment (see Supplemental Materials 1.2 for picture codes). Participants practiced applying the VIEW or REAPPRAISE instructions and responding to the self-report questions used to assess subjective affect and appraisals. To ensure that participants focused on positive goal reappraisal, they wrote down what they had thought about to feel more positive after each reappraisal practice trial and then read standardized examples illustrating how this could be done.

Each trial (see Figure 1) began with a fixation cross (1,000 ms), followed by a picture (6,000 ms), and then scales assessing affect and appraisals in a fixed order (until response).

In one randomly selected trial from each valence category in each REAPPRAISE block, participants were asked to briefly write about how they reappraised the depicted situation. We excluded seven

Figure 1
Single Trial in Study 1



Note. Stimuli were selected from the International Affective Picture System (IAPS; Lang et al., 2005), the Nencki Affective Picture System (NAPS; Marchewka et al., 2014), the Geneva Affective Picture Database (GAPED; Dan-Glauser & Scherer, 2011), and the Emotional Picture Set (EmoPicS; Wessa et al., 2010).

participants whose responses to at least three of the six questions either lacked comprehensible text, indicated that they had not attempted to apply reappraisal, or otherwise suggested noncompliance with the instructions. In addition, three attention check questions were presented at irregular intervals during the study (see Supplemental Materials 1.5). All participants correctly answered at least two out of three attention check questions; therefore, no one was excluded based on this criterion.

The postexperiment questionnaire included the short form of the Marlowe–Crowne Social Desirability Scale (Reynolds, 1982) and items assessing the difficulty of the self-report questions and the overall task (see Supplemental Materials 1.6).

Self-Report Measures

Subjective positive and negative affect were assessed using two unipolar items that asked how negative and how positive participants felt either while viewing the picture (the VIEW condition) or after trying to change their thinking (the REAPPRAISE condition). Responses were recorded on 9-point Likert scales ranging from (*no negative feelings*) to (*strong negative feelings*) and (*no positive feelings*) to (*strong positive feelings*), respectively.

Appraisals of the depicted situations were assessed using six items phrased to complete the sentence “While imagining I was in the depicted situation, I thought that …” (the VIEW condition) or “After trying to change how I think about the depicted situation, I thought that …” (the REAPPRAISE condition). Responses were recorded on a 9-point Likert scale ranging from (*strongly disagree*) to (*strongly agree*). The items targeted five appraisal dimensions: relevance (“… this situation matters to me”), other-accountability (“… someone is responsible for this situation”), current certainty (“… it is clear to me what is going on in this situation”), controllability (“… I can change this situation”), and congruence. Congruences was assessed using two items—one for current congruence (“… this situation is in line with what I want”) and another for outlook congruence (“… this situation will turn out the way I want”) that was included to capture the effects of the reappraisal tactic of changing future consequences (McRae et al., 2012).

Statistical Analyses

We did not use the preregistered within-subjects moderated mediation analyses to avoid implying causal inferences that our data do not fully support (Rohrer et al., 2022). Instead, we analyzed the effects of reappraisal on subjective positive and negative affect (Research Question 1) using separate LMMs, with condition (VIEW and REAPPRAISE) and valence category (neutral, negative, and positive) as fixed factors. We applied deviation coding so that the regression coefficients reflect differences from the baseline level of each factor, averaged across the levels of the other factors in the model. The models also included random intercepts for each participant and stimulus, as well as random slopes for the effects of stimulus and condition by participant. Random slopes for stimulus–condition interactions were omitted to fix model convergence issues, in line with recommendations by Barr et al. (2013).

The involvement of appraisal shifts in reappraisal-related affect changes (Research Question 2) was analyzed using separate LCSMs for subjective positive and negative affect (see Supplemental Materials 1.7). For each valence category, we averaged the appraisal and affect scores within each condition and constructed latent variables representing the change from the VIEW to the REAPPRAISE condition for both appraisals and affect. In each model, latent change in affect was regressed on latent changes in appraisals, with valence category included as a moderating variable. We interpreted significant regression coefficients (β) as indicating the involvement of a given appraisal shift in affect change, and significant differences between these coefficients across valence categories as indicating that the role of the shift varied by valence category. The p values for all tests within a single model were corrected using the false discovery rate (FDR) correction algorithm (Benjamini & Yekutieli, 2001).

Analyses were performed using R 4.3.0 (R Core Team, 2023) with the RStudio platform 2023.12.1 (RStudio Team, 2023) and the tidyverse (Wickham et al., 2019), lavaan (Rosseel, 2012), and rstatix (Kassambara, 2023) packages.

Results and Discussion

Reappraisal Effects on Affect Ratings

The results of the LMMs investigating Research Question 1 are presented in Table 1 and illustrated in Figure 2, where the arrows indicate changes from the VIEW to REAPPRAISE condition, representing both mean positive affect (x-axis) and negative affect (y-axis) scores. Indicating successful affect elicitation, negative pictures were rated significantly more negatively and less positively than neutral pictures. Positive pictures were rated less negatively and more positively than neutral pictures.

The condition main effects confirmed that applying positive goal reappraisal resulted in higher positive and lower negative affect. Significant interactions between condition and valence category indicated that reappraisal effects varied across valence categories (see Supplemental Materials 1.8 for mean affect ratings). The increase in positive affect was most pronounced for negative pictures (marginal mean difference $d = 1.88$, $p < .001$), followed by neutral ($d = 1.22$, $p < .001$) and positive pictures ($d = 0.38$, $p < .001$). Similarly, the decrease in negative affect was largest for negative pictures ($d = 1.74$, $p < .001$), followed by neutral ($d = 0.47$, $p < .001$) and positive pictures ($d = 0.18$, $p = .003$). These results replicate previous findings

Table 1

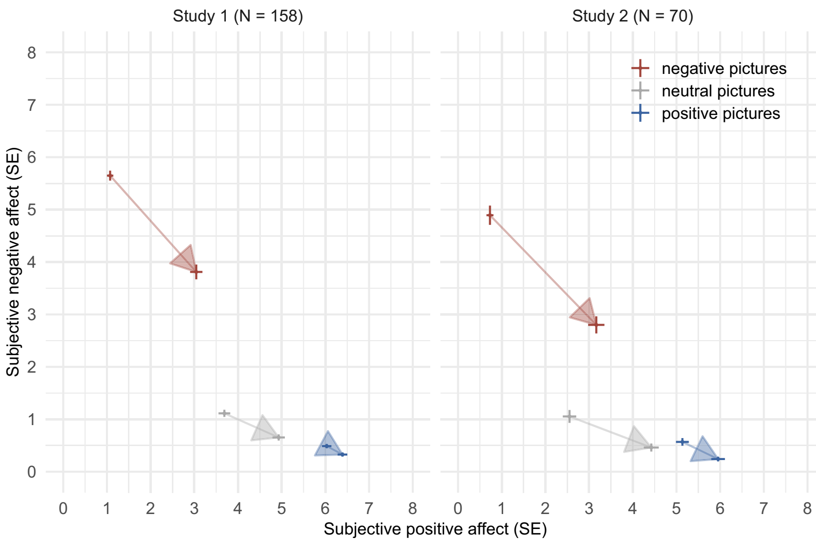
Experimental Effects on Subjective Affect in Study 1 (Left Column) and Study 2 (Right Column)

	Study 1				Study 2			
	Positive affect		Negative affect		Positive affect		Negative affect	
Model term	β	p	β	p	β	p	β	p
Mean (intercept)	4.19	**<.001**	2.01	**<.001**	3.63	**<.001**	1.67	**<.001**
Negative (vs. neutral)	−2.25	**<.001**	3.85	**<.001**	−1.53	**<.001**	3.08	**<.001**
Positive (vs. neutral)	1.91	**<.001**	−0.48	**<.001**	2.04	**<.001**	−0.35	**.02**
REAPPRAISE (vs. VIEW)	1.16	**<.001**	−0.80	**<.001**	1.71	**<.001**	−0.98	**<.001**
REAPPRAISE × Negative	0.66	**<.001**	−1.27	**<.001**	0.55	**<.001**	−1.47	**<.001**
REAPPRAISE × Positive	−0.84	**<.001**	0.30	**<.001**	−1.06	**<.001**	0.26	**<.001**
σ^2 (error variance)	2.40		2.29		1.93		1.27	
τ_{00} (random intercept variance)								
Stimulus	0.35		0.44		0.24		0.25	
Participant	1.50		0.42		1.03		0.51	
τ_{11} (random slope variance)								
Participant × Negative	1.35				0.53		0.87	
Participant × Positive	0.56				0.47		0.16	
Participant × REAPPRAISE	0.75		0.28		0.77		0.35	
ρ_{01} (random intercept–slope correlation)								
Participant × Negative	−0.55				−0.47		0.50	
Participant × Positive	−0.50				−0.07		−0.77	
Participant × REAPPRAISE	0.40		0.30		0.51		−0.32	
ICC (intra-class correlation)	0.50		0.29		0.46		0.46	
N								
Participant	158		158		70		70	
Stimulus	144		144		84		84	
Observations	15,168		15,168		5,880		5,880	
Marginal R^2/conditional R^2	0.41/0.70		0.56/0.68		0.45/0.71		0.54/0.75	

Note. Results of four separate LMMs with condition (VIEW, REAPPRAISE) and valence category (neutral, negative, positive) as fixed factors. β = unstandardized beta. All statistically significant p values (<.05) are in bold. Some coefficients for negative affect are missing in Study 1 due to model convergence issues. LMMs = linear mixed models.

Figure 2

Means and Standard Errors of Subjective Positive and Negative Affect Across Valence Categories in Study 1 (Left) and Study 2 (Right)



Note. Subjective positive and negative affect was measured using two separate 9-point Likert scales, with higher scores indicating stronger affect. The beginning of each arrow depicts the mean negative affect (y-axis) and mean positive affect (x-axis) in the VIEW condition, and the end of each arrow depicts the respective mean in the REAPPRAISE condition. SE = standard error. See the online article for the color version of this figure.

regarding reappraisal effects on positive affect for negative and positive stimuli, and negative affect for negative stimuli (e.g., Giuliani et al., 2008; McRae et al., 2012; Samson et al., 2014; Waugh et al., 2016).

Appraisal Shifts and Affect Ratings

The results of the LCSMs investigating Research Question 2 are illustrated in Figure 3, which depicts the associations between latent appraisal shifts and latent changes in subjective affect (see Supplemental Materials 1.9 for additional LCSM coefficients). Consistent with previous research (A. Uusberg et al., 2023), we observed significant shifts in appraisals across all valence categories (see Supplemental Materials 1.8 and 1.10), many of which were significantly related to changes in subjective positive and negative affect between the VIEW and REAPPRAISE conditions. Based on R^2 values, appraisal shifts explained 38.7%, 24.1%, and 24.2% of the increase in subjective positive affect, as well as 47.3%, 19.1%, and 9.2% of the decrease in subjective negative affect in response to negative, neutral, and positive pictures, respectively.

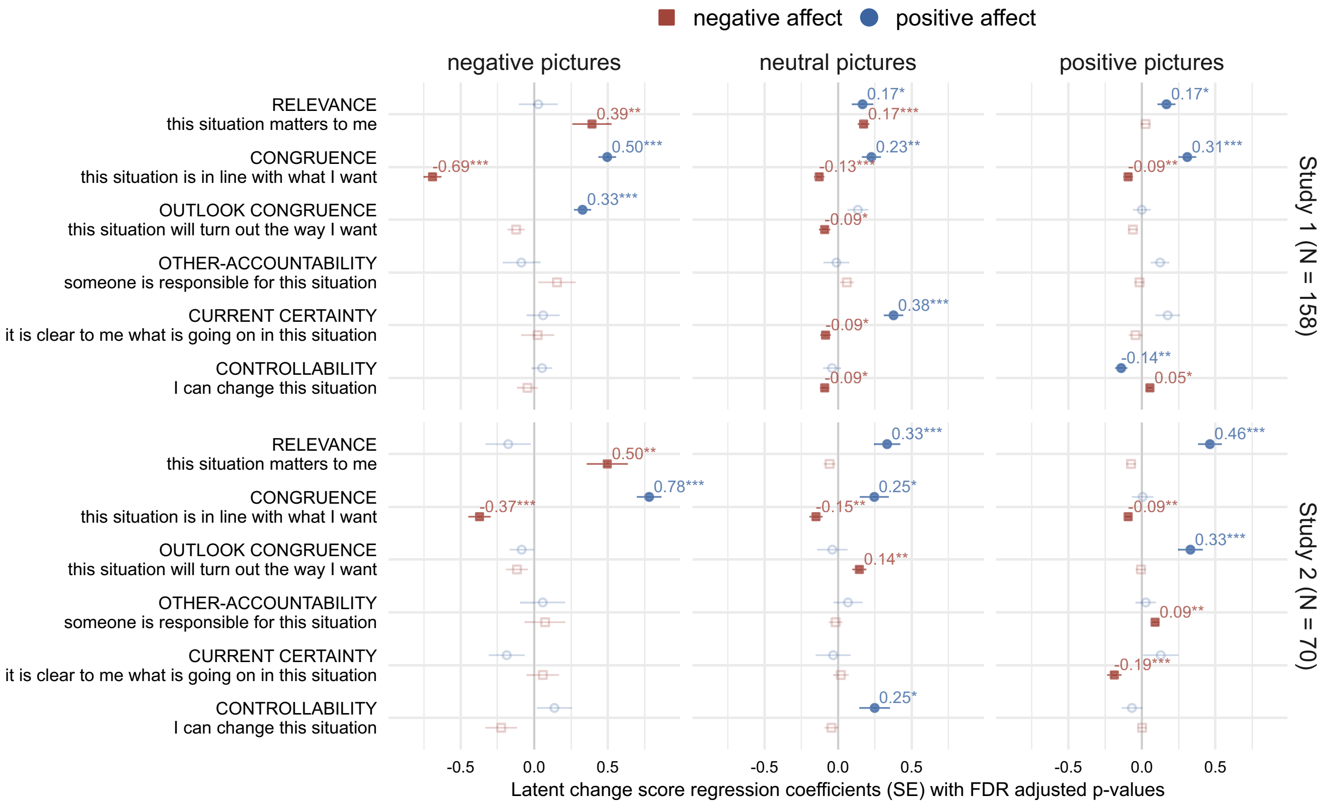
Consistent with prior research (A. Uusberg et al., 2023), we observed the strongest effects for shifts in congruence, which were consistently associated with higher positive affect and lower negative affect across all valence categories. Effects were most pronounced for negative pictures, where increased congruence was related to higher positive affect (unstandardized $\beta = 0.50$) and lower negative affect ($\beta = -0.69$). Smaller but similar effects emerged for neutral (positive affect: $\beta = 0.23$; negative affect: $\beta = -0.13$) and positive pictures (positive affect: $\beta = 0.31$; negative affect: $\beta = -0.09$). Although outlook congruence played a comparatively smaller role, increased outlook congruence was also significantly associated with higher positive affect for negative pictures ($\beta = 0.33$) and lower negative affect for neutral pictures ($\beta = -0.09$).

Changes in relevance were also associated with changes in subjective affect and the pattern varied by valence category. For negative pictures, decreased relevance was associated with lower negative affect ($\beta = 0.39$). For neutral pictures, increased relevance was associated with higher positive affect ($\beta = 0.17$), whereas decreased relevance was related to lower negative affect ($\beta = 0.17$). For positive pictures, increased relevance was related to higher positive affect ($\beta = 0.17$). This pattern is consistent with the idea that relevance is reflected in affective arousal rather than affective valence (Moors et al., 2013; Yeo & Ong, 2024). Consequently, changes in relevance may serve quantitative (i.e., changing emotion intensity)—rather than qualitative (i.e., changing emotion type)—emotion regulation goals (Kreibig et al., 2023).

Whereas only congruence, outlook congruence, and relevance played a significant role for negative pictures, additional appraisal shifts were involved in reappraisal of neutral and positive pictures. For neutral pictures, increased current certainty was associated with higher positive affect ($\beta = 0.38$) and lower negative affect ($\beta = -0.09$). Additionally, an interesting dissociation emerged for controllability. Affective improvements are usually associated with increased controllability (Skinner, 1996; A. Uusberg et al., 2023), and consistent with this, we found that lower negative affect was associated with increased controllability for neutral pictures ($\beta = -0.09$). However, for positive pictures, higher positive affect ($\beta = -0.14$) and lower negative affect ($\beta = 0.05$) were associated with decreased controllability. This suggests that in positive contexts, relinquishing control and adopting a more passive stance toward future developments may be beneficial. Such a shift might enhance emotional presence and

Figure 3

Associations Between Latent Appraisal Shifts and Latent Changes in Subjective Affect



Note. LCSM coefficients representing the associations between a given appraisal shift and changes in subjective positive affect (blue points) and subjective negative affect (red points) in response to negative (left column), neutral (middle column), and positive (right column) pictures in Study 1 (top panel) and Study 2 (bottom panel). Numbers represent unstandardized regression coefficients derived from the LCSMs. Dark filled points indicate statistically significant coefficients. *SE* = standard error; LCSM = latent change score modeling; FDR = false discovery rate. See the online article for the color version of this figure. Stars denote the levels of FDR adjusted *p* values: * < .05. ** < .01. *** < .001.

reduce cognitive load, allowing for fuller engagement with the positive aspects of the situation. More broadly, these findings indicate that reappraisal mechanisms may be context-dependent, engaging a broader range of dimensions in neutral and positive contexts than in negative ones.

Taken together, the findings from Study 1 indicate that positive goal reappraisal increased subjective positive affect and decreased subjective negative affect in response to negative, neutral, and positive pictures, suggesting potential benefits across all valence categories. The study also showed that several appraisal shifts accounted for a substantial portion of the effects of reappraisal on self-reported affect. Shifts in congruence and relevance were consistently related to reappraisal effects across all valence categories, whereas associations with certainty and controllability varied by valence category.

Study 2: Laboratory Experiment

In Study 2, we aimed to replicate and extend the findings of Study 1 while also addressing some of its limitations. First, the parallel sets used in Study 1 were equivalent on most, but not all, affect and appraisal ratings (see Supplemental Materials 1.3). We therefore created new parallel stimulus sets for Study 2 using ratings from Study 1. Second, self-reports of affect may have been biased due to demand characteristics, especially since the reappraisal instruction made it clear what was being studied. To address this limitation, we conducted Study 2 in a laboratory setting, using facial muscle reactivity and event-related potentials to measure nonexperiential components of affect (Mauss & Robinson, 2009), in addition to self-reported affect and appraisals.

To assess the expressive component of affect, we analyzed the reactivity of two facial muscles—*zygomaticus major* and *corrugator supercilii*—which are involved in the display of positive and negative affect and are commonly used to assess emotional responding (e.g., Larsen & Norris, 2009; Larsen et al., 2003) as well as emotion regulation (e.g., Kreibig et al., 2023; Schönfelder et al., 2014). Previous research has shown that positive goal reappraisal can increase *zygomaticus* and decrease *corrugator* responses to negative stimuli (e.g., Kreibig et al., 2023; vanOyen Witvliet et al., 2010). An increase in *zygomatic* response has also been found with positive stimuli (e.g., Li et al., 2018). To our knowledge, no study has employed facial muscle reactivity to assess positive goal reappraisal across negative, neutral, and positive stimuli simultaneously.

We also analyzed the amplitudes of the late positive potential (LPP), an ERP component associated with affective arousal and motivated attention (Cuthbert et al., 2000; Hajcak et al., 2010;

Olofsson et al., 2008; Schupp et al., 2006). Reappraisal aimed at reducing negative affect can decrease LPP amplitudes (e.g., Hajcak & Nieuwenhuis, 2006; Hajcak et al., 2010), whereas the effects of positive goal reappraisal are less clear. For instance, Liu et al. (2019) and Cheng et al. (2023) found an increase in LPP amplitudes when participants used reappraisal to increase positive affect in response to positive stimuli, while Krompinger et al. (2008) found no effect. By incorporating ERP markers of affective experience, we can gain further insights into how positive goal reappraisal impacts the processing of negative, neutral, and positive stimuli.

Method

Participants

As preregistered, we recruited 70 Estonian-speaking adult volunteers (52 female, 17 male, one other; age range 18 and 51 years, M = 26.2, SD = 7.9), primarily university students, to achieve 80% power to detect within-subject differences of d = 0.34. Participants were not taking medication that could affect their emotional experience and were not undergoing psychotherapy. Due to artefactual signals, two participants were excluded from EMG analyses (see the EEG and EMG Recording and Preprocessing section; final N = 68, 50 female, 17 male, one other; M = 25.9 years, SD = 7.9), and six from EEG analyses (final N = 64, 51 female, 12 male, one other; M = 25.9 years, SD = 7.9). Prior to the experiment, all participants provided informed consent. No monetary compensation was offered. Participants received feedback on the personality questionnaire. Undergraduate psychology students at the University of Tartu received research participation credit, while all other participants had the opportunity to enter a draw for a €30 bookstore gift card.

Stimuli

Two new parallel sets of 14 negative, 14 neutral, and 14 positive pictures were created from the stimuli used in Study 1 by incorporating the mean affect and appraisal ratings from Study 1 into the features used to balance the valence categories and parallel sets. All other features and set-balancing procedures remained identical to those in Study 1. See Supplemental Materials 2.1–2.3 for picture codes, descriptive statistics of the stimulus sets, and picture-level data collected in this study.

Procedure and Design

Study 2 consisted of a preexperiment questionnaire on the https://formr.org platform, a laboratory experimental task, and a postexperiment questionnaire on the https://formr.org platform. Data were collected between September and December 2023. See Supplemental Materials 2.4 for the list of the pre- and post-experiment questionnaires.

Participants completed the 2 (condition: VIEW, REAPPRAISE) by 3 (valence category: negative, neutral, positive) within-subject experiment, which was identical to Study 1 except for the following differences. All written instructions and self-report items were translated into Estonian. The number of pictures per block was reduced from 16 to 14. Each participant thus saw 84 affective pictures, with one randomly selected parallel set of 14 negative, 14 neutral, and 14 positive pictures presented in each condition. The stimuli were presented full screen on a 19-in. liquid crystal display monitor (screen resolution: 1,024 × 768) at a 80-cm distance in a dimly lit and electrically shielded room.

During practice trials, participants summarized the written instructions and described their thoughts aloud while viewing or reappraising the pictures, rather than writing them down. The experimenter verbally presented standardized examples illustrating potential ways to reappraise the depicted situations. The experimenter encouraged participants to ask questions, and provided corrective feedback, if necessary.

The structure of a single trial was altered slightly to accommodate psychophysiological measures (see Figure 4). Each trial began with an inter-trial interval for a random duration between 750 and 1,250 ms (M = 1,000 ms), followed by a fixation cross for a random duration between 750 and 1,250 ms (M = 1,000 ms). A fixation cross was followed by a picture for 6,000 ms and a blank screen for 500 ms. The blank screen was followed by the scales assessing affect and appraisals in a fixed order (until response).

Unlike Study 1, Study 2 did not include free response questions and attention check questions. Instead, after each block, participants were asked to rate how often they managed to either view or reappraise the situations that were depicted in the pictures, using a 9-point Likert scale ranging from (*almost never*) to (*almost always*). After each REAPPRAISE block, participants also rated the extent to which they either reconsidered what was going on in the situations (i.e., reconstrual; A. Uusberg et al., 2019) or re-evaluated how much the situations mattered to them and why (i.e., repurposing; A. Uusberg et al., 2019) on a 9-point Likert scale.

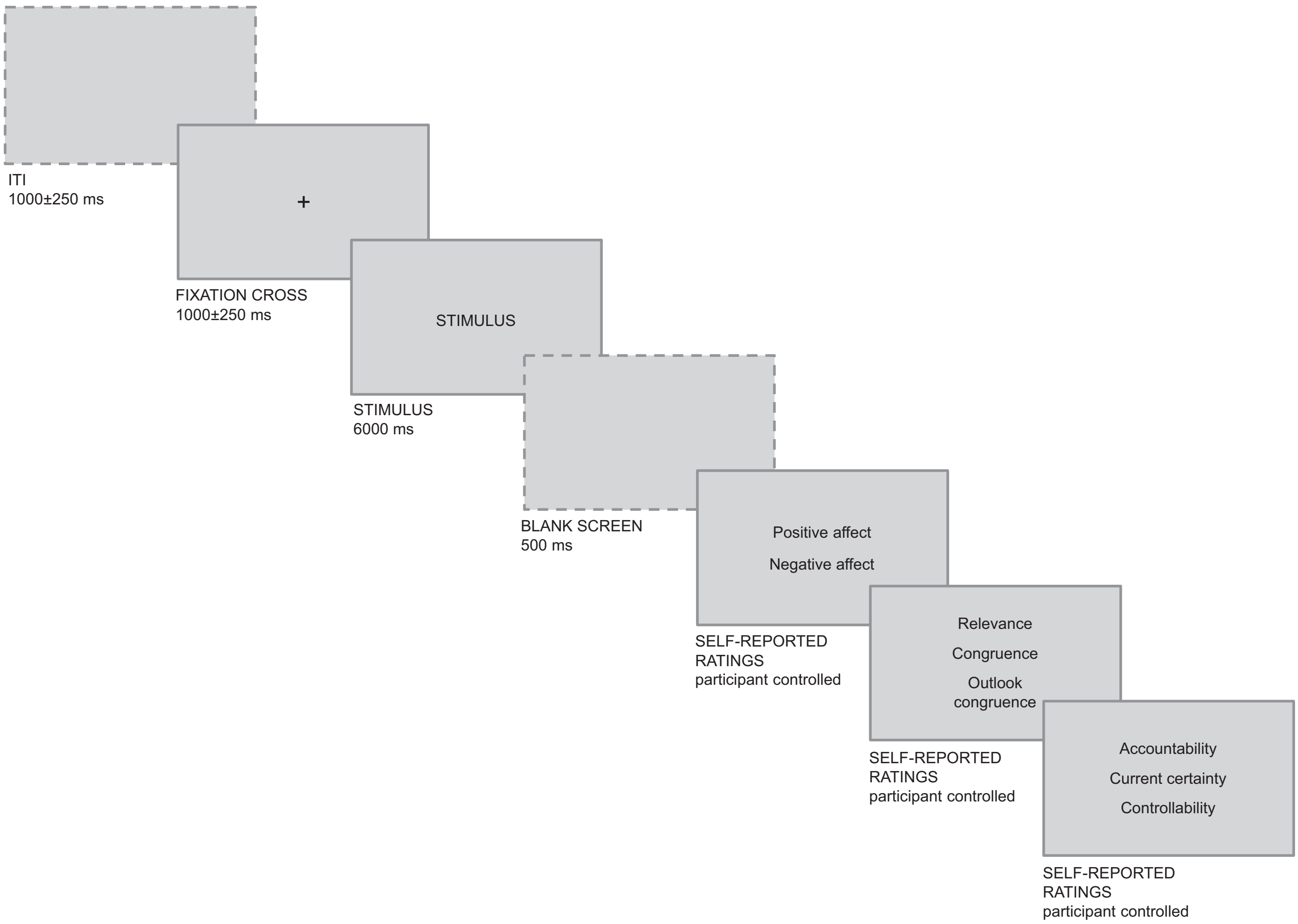
After the experimental task, participants entered their demographic data and completed the postexperiment questionnaire that included questions about the experimental task like the ones asked in Study 1 (see Supplemental Materials 2.5). To verify adherence to the instructions, participants were also asked to recall and provide short free responses describing how they reappraised the situations depicted in one negative, one neutral, and one positive picture. None of the participants were excluded, as they all provided comprehensible text and indicated adherence to the instructions.

EEG and EMG Recording and Preprocessing

Continuous EEG was recorded using a BioSemi ActiveTwo system from 32 scalp locations, both earlobes, and bipolar horizontal ocular locations, with the common mode sense/driven right leg reference scheme and a 512 Hz sampling rate. Offline processing was performed using EEGLAB v13.6.5b (Delorme & Makeig, 2004) on MATLAB R2023a (MathWorks, 2023).

All scalp EEG channels were re-referenced to linked earlobes. Noisy scalp channels and epochs were first removed using EEGLAB automatic rejection algorithms *rejcan* and *autorej* with default parameters. Independent component analysis was then applied to computationally remove ocular artifacts. The infomax algorithm was trained on 4-s (−1000 to 3,000 ms) epochs of 1 Hz high-pass filtered data. Manually identified components corresponding to blinks and eye movements were removed before reconstructing the unfiltered continuous data by multiplying the raw data with the independent component analysis mixing matrix using the remaining components. The independent component analysis-pruned data were then high-pass filtered at 0.1 Hz, low-pass filtered at 30 Hz, and segmented from −500 ms before stimulus onset to 6,000 ms (i.e., stimulus offset).

Figure 4
Single Trial in Study 2



Note. Stimuli were selected from the International Affective Picture System (IAPS; Lang et al., 2005), the Nencki Affective Picture System (NAPS; Marchewka et al., 2014), the Geneva Affective Picture Database (GAPED; Dan-Glauser & Scherer, 2011), and the Emotional Picture Set (EmoPicS; Wessa et al., 2010). The dashed line indicates the parts of the trial structure that differed from Study 1. ITI = inter-trial interval.

The mean of the 200 ms prestimulus voltage was removed as a baseline. Epochs were discarded as artifacts if the voltage deviated more than 150 μV from the baseline. If more than 2% of segments were to be removed due to a single channel, the channel was removed instead. Six participants were excluded from analysis due to losing more than 50% of trials in any of the two condition by three valence category design cells. In the final sample, on average 91.80% (SD = 9.46%; range = 65.48%–100%) of the trials were retained. This translates to at least 9.17 and, on average, 12.85 trials per each mean analyzed in this study, exceeding the recommended minimum of eight trials for reliable LPP estimates (Moran et al., 2013). The trial retention rate was independent of condition, valence category, and their interaction ($ps > .05$, rANOVA).

The LPP was quantified as the mean amplitude across a central-parietal electrode cluster (Cz, CP1, CP2, Pz) from 250 to 6,000 ms after stimulus onset and divided into two time windows: an early LPP (250–1,500 ms) and a late LPP (1,500–6,000 ms). These decisions were guided by previous studies (reviewed in Hajcak et al., 2010; Olofsson et al., 2008; Schupp et al., 2006) and by the spatial and temporal distribution of the valence category main effect in the present study. We simplified our preregistered strategy by omitting a separate frontal-central electrode cluster, as it showed dynamics similar to the central-parietal cluster. Likewise, we did not distinguish between the P3 and Early Slow Wave components because they were not clearly separable in the grand-average ERP waveform (see Supplemental Materials 2.6). The Spearman–Brown split-half reliability coefficients for the early and late LPP mean amplitudes were .59 and .35, respectively. The more precise time course of affective modulation was analyzed using a mass-univariate approach (see the Statistical Analyses section).

Facial muscle reactivity was measured from the *corrugator supercilii* with two electrodes placed 1 cm apart at equal distance above the left eyebrow, and from the *zygomaticus major* with two electrodes placed 2 cm apart along the imaginary line from the left corner of the mouth to the ear. For each facial muscle, the signals from the two electrodes were subtracted, high-pass filtered at 20 Hz, notch filtered between 48 and 52 Hz, rectified by taking the square root of each time point, and finally smoothed by computing a moving average within a

125-ms window. After segmenting the data from −750 ms to 6,000 ms relative to stimulus onset, the prestimulus values were subtracted as a baseline from each trial. The signal was then converted to *z*-scores for each muscle and each participant. Segments were removed as artifacts if their values deviated more than five within-subject *SD* from the segment baseline at any point. Two participants were excluded from analysis due to losing more than 50% of trials in any of the two condition by three valence category design cells. In the final sample, on average 93.05% ($SD = 3.35\%$; range = 84.52%–98.81%) of the trials were retained. This translates to at least 11.83 and, on average, 13.03 trials per each mean analyzed in this study, exceeding the recommended minimum of eight trials for reliable EMG estimates (Larsen et al., 2003). The trial retention rate depended on the interaction between condition and valence category ($p = .006$; rANOVA). Specifically, on average, fewer single trials were retained for positive pictures in the REAPPRAISE condition ($M = 12.32$; $SE = 0.16$; $ps < .001$) compared to other types of pictures in the VIEW and REAPPRAISE conditions (*M* range = 12.91–13.32; *SE* range = 0.10–0.14; $ps > .19$, rANOVA). However, since the differences are small and each design cell included more than eight trials, we do not consider this a major concern.

EMG reactivity was calculated as the mean baseline-corrected activity across the entire period from 0 to 6,000 ms. The Spearman–Brown split-half reliability coefficients for average *zygomaticus major* and *corrugator supercilii* reactivity were .81 and .86, respectively. The more precise time course of affective modulation was analyzed using a mass-univariate approach (see the Statistical Analyses section), instead of the preregistered strategy of analyzing six subsequent 1,000 ms averages.

Statistical Analyses

Following Study 1, we used LMMs to investigate the effects of condition and valence category on affect ratings, EMG reactivity, and LPP amplitudes (Research Question 1). The fixed and random effects structure of the models was identical to that used in Study 1. In the LPP model, time window (250–1,500 ms, 1,500–6,000 ms) was included as an additional fixed factor.

We also used a mass-univariate approach (Groppe et al., 2011) to investigate the timing of condition and valence category effects on EMG and LPP by conducting separate *t*-tests or analyses of variance for each time point and correcting the resulting *p* values using the FDR algorithm (Benjamini & Yekutieli, 2001).

As in Study 1, the involvement of appraisal shifts in reappraisal-related changes in different affect measures (Research Question 2) was analyzed using LCSMs (see the Statistical Analyses section in Study 1). We opted for LCSMs instead of the preregistered linear regression approach, as LCSMs avoids reliance on difference scores, which can amplify measurement error.

Results and Discussion

Reappraisal Effects on Affect Ratings

The results of the LMMs investigating Research Question 1 are presented in Table 1 and illustrated in Figure 2. As in Study 1, the main effects of valence category indicated successful affect elicitation. Negative pictures were rated more negatively and less positively than neutral pictures and positive pictures were rated less negatively and more positively than neutral pictures. The condition main effects confirmed that applying positive goal reappraisal resulted in increased positive and decreased negative affect. Significant interactions between condition and valence category indicated that reappraisal effects varied across valence categories (see Supplemental Materials 2.7 for mean affect ratings). As in Study 1, the increase in positive affect was most pronounced for negative pictures (marginal mean difference $d = 2.43$, $p < .001$), followed by neutral ($d = 1.88$, $p < .001$) and positive pictures ($d = 0.82$, $p < .001$). Similarly, the decrease in negative affect was largest for negative pictures ($d = 2.04$, $p < .001$), followed by neutral ($d = 0.57$, $p < .001$) and positive pictures ($d = 0.32$, $p < .001$).

Reappraisal Effects on EMG Reactivity

Table 2 shows the LMM results for *zygomaticus* and *corrugator* reactivity. As expected, *zygomaticus* reactivity was highest for positive pictures, followed by neutral pictures, and then negative pictures. *Corrugator* reactivity was highest for negative pictures, followed by neutral pictures, and then positive pictures. These findings align with prior research showing that *zygomaticus* and *corrugator* can serve as valuable markers of positive and negative affect, respectively (e.g., Kreibig et al., 2023; Larsen & Norris, 2009; Larsen et al., 2003; Schönfelder et al., 2014).

The efficacy of reappraisal was confirmed by condition main effects on both muscles. Specifically, *zygomaticus* reactivity was higher and *corrugator* reactivity was lower in the REAPPRAISE condition compared to the VIEW condition. Further distinctions

Table 2

Experimental Effects on Zygomaticus Major and Corrugator Supercilii Reactivity

Model term	Zygomaticus major β	Zygomaticus major *p*	Corrugator supercilia β	Corrugator supercilia *P*
Mean (intercept)	0.09	**<.001**	−0.11	**.001**
Negative (vs. neutral)	−0.10	**<.001**	0.35	**<.001**
Positive (vs. neutral)	0.24	**<.001**	−0.29	**<.001**
REAPPRAISE (vs. VIEW)	0.13	**<.001**	−0.09	**<.001**
REAPPRAISE × Negative	−0.13	**<.001**	0.08	.06
REAPPRAISE × Positive	0.02	.52	0.24	**<.001**
σ^2 (error variance)	0.29		0.41	
τ_{00} (random intercept variance)				
Stimulus	0.00		0.01	
Participant	0.04		0.06	
τ_{11} (random slope variance)				
Participant × Negative	0.01		0.06	
Participant × Positive	0.04		0.03	
ρ_{01} (random intercept–slope correlation)				
Participant × Negative	−0.63		0.03	
Participant × Positive	0.87		0.53	
ICC (intra-class correlation)	0.15		0.18	
N				
Participant	68		68	
Stimulus	84		84	
Observations	5,315		5,315	
Marginal R^2/conditional R^2	0.07/0.21		0.13/0.28	

Note. Results of two separate LMMs with condition (VIEW, REAPPRAISE) and valence category (neutral, negative, positive) as fixed factors. β= unstandardized beta. All statistically significant *p* values (<.05) are in bold. LMMs = linear mixed models.

emerged through interaction effects between condition and valence category (see Supplemental Materials 2.7 for mean EMG responses). For negative pictures, reappraisal decreased *corrugator* reactivity (d = 0.12, p < .001), without a corresponding increase in *zygomaticus* reactivity (d = 0.03, p = .18). Mass-univariate analyses revealed that the reappraisal effect on *corrugator* reactivity was significant between 1,363 and 1,488 ms, 2037 and 2,263 ms, and from 4,713 ms until the end of stimulus presentation (excluding changes in statistical significance lasting less than 100 ms; see Figure 5). Meanwhile, the reappraisal effect remained absent for *zygomaticus* reactivity throughout the entire stimulus presentation.

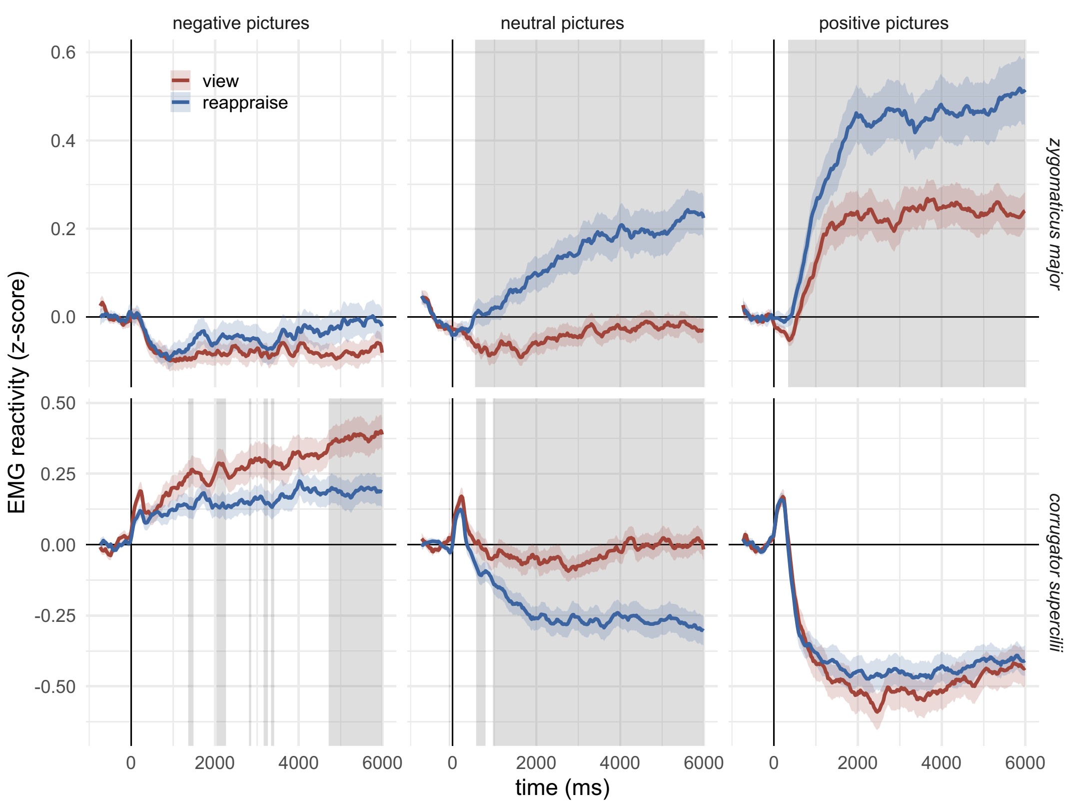
Lower *corrugator* reactivity in response to negative stimuli aligns with previous studies on positive goal reappraisal (e.g., Kreibig et al., 2023; vanOyen Witvliet et al., 2010). However, the absence of change in *zygomaticus* reactivity contradicts these findings. This may reflect the longer time required to produce a qualitative change in the expressive component of affect (i.e., increasing positive affect in a negative situation) compared to a quantitative change (i.e., reducing negative affect in a negative situation). Kreibig and Gross (2024) note that qualitative shifts in facial muscle reactivity may follow quantitative shifts. This suggests that any effect reappraisal may have had on *zygomaticus* reactivity in response to negative pictures would have emerged only after changes in *corrugator* reactivity, which became apparent only in the final second of stimulus presentation. It is therefore possible that presenting the negative picture alongside the reappraisal instruction for longer than 6 s would have allowed the full expressive response to unfold.

Aligning with prior research (e.g., Li et al., 2018), reappraisal of neutral pictures increased *zygomaticus* reactivity (d = 0.16, p < .001) and decreased *corrugator* reactivity (d = 0.20, p < .001). The increase in *zygomaticus* reactivity was significant from 537 ms until the end of stimulus presentation. The decrease in *corrugator* reactivity was significant between 563 and 787 ms, as well as from 963 ms until the end of stimulus presentation (see Figure 5). These findings underscore the potential of using reappraisal in neutral contexts to improve affect and, in turn, support overall well-being (Lyubomirsky et al., 2005; McRae & Mauss, 2016).

For positive pictures, reappraisal increased *zygomaticus* reactivity (d = 0.19, p < .001) but did not reduce *corrugator* reactivity (d = 0.04, p = .15). The increase in *zygomaticus* reactivity was significant from 338 until the end of stimulus presentation. For *corrugator*, the reappraisal effect remained insignificant throughout the stimulus presentation (see Figure 5). The increase in *zygomaticus* reactivity following positive goal reappraisal is consistent with prior studies

Figure 5

Time Courses of Condition Effects on Facial Muscle Reactivity Across Valence Categories



Note. The temporal dynamics of zygomaticus major (top panel) and corrugator supercilii (bottom panel) reactivity in response to negative (left column), neutral (middle column), and positive (right column) pictures in the VIEW (red line) and REAPPRAISE (blue line) conditions. Gray areas depict significant differences between the VIEW and REAPPRAISE conditions (mass-univariate t-test, FDR-corrected, $p < .05$, $N = 68$). EMG = electromyography; FDR = false discovery rate. See the online article for the color version of this figure.

(e.g., Li et al., 2018). The absence of a decrease in *corrugator* reactivity suggests a potential floor effect, implying that participants might not have expressed negative affect strongly enough to further decrease it.

Additionally, consistent with prior research (e.g., Davidson & Irwin, 1999; Larsen et al., 2003; Mauss et al., 2005), we found statistically significant but modest within-subject correlations between subjective positive affect and *zygomaticus* reactivity (r = .16, p < .001), and between subjective negative affect and *corrugator* reactivity (r = .18, p < .001). This suggests both shared and unique variance between experiential and expressive components of affect. Subjective and facial expression measures aligned most clearly for neutral pictures, where both changed significantly. For positive pictures, both subjective positive affect and *zygomaticus* reactivity increased, while a floor effect emerged for *corrugator* reactivity but not for subjective negative affect. For negative pictures, reappraisal effects were more pronounced in subjective affect, which showed changes in both positive and negative affect, while facial expression changes were limited to a late decrease in *corrugator* reactivity.

Given the block design of the experiment, the effects we observed on poststimulus EMG reactivity could have been modulated by experimental effects on prestimulus baseline values, which were subtracted from the poststimulus values. To examine this, we tested for differences in prestimulus baselines between the VIEW and REAPPRAISE blocks (see Supplemental Materials 2.8). We found that *zygomaticus major* baselines were higher and *corrugator supercilii* baselines were lower in the REAPPRAISE blocks compared to the VIEW blocks. Because reappraisal influenced both pre- and poststimulus EMG reactivity in the same direction, the EMG reactivity effects we observed are unlikely to be artifacts of baseline differences. Instead, they likely reflect conservative estimates of reappraisal effects, which would have been even larger without baseline correction.

Reappraisal Effects on LPP Amplitudes

Table 3 shows the LMM results for LPP amplitudes. A significant valence category main effect confirmed that both negative and positive pictures elicited larger LPP responses than neutral pictures, consistent with the role of the LPP as a marker of motivated attention sensitive to emotional intensity (e.g., Hajcak et al., 2010; Olofsson et al., 2008; Schupp et al., 2006). In line with previous findings (Hajcak & Nieuwenhuis, 2006; Weinberg & Hajcak, 2010), LPP amplitudes did not differ significantly between positive and negative pictures (p = .83). According to the mass-univariate analyses, LPP amplitudes were significantly larger for negative and positive pictures compared to neutral pictures from 288 ms until the end of stimulus presentation (see Figure 6, Panel B).

Previous research has shown that reappraisal of positive pictures either increases (Cheng et al., 2023; Liu et al., 2019) or does not affect (Krompinger et al., 2008) LPP amplitudes. In our study, LPP amplitudes were higher in the REAPPRAISE condition than in the VIEW condition across all valence categories (see Supplemental Materials 2.7 for mean LPP responses). Mass-univariate analyses revealed that this effect emerged relatively early, between 263 and 1,013 ms (see Figure 6, panel A). A plausible interpretation is that this valence-unspecific LPP increase reflects enhanced attentional allocation to the stimuli during the generation of alternative appraisals. This interpretation is consistent with the relatively early timing of the effect, which overlaps with the P3 component associated with intentional attention (Olofsson et al., 2008). Supporting this view, prior studies have shown that attending to, versus not attending to, both emotional and neutral stimuli can indeed modulate LPP amplitudes (e.g., Hajcak et al., 2013; H. Uusberg et al., 2016). Moreover, we did not find an interaction between condition and valence category (see Table 3), as would be expected if reappraisal-related differences in LPP amplitudes reflected changes in arousal. The early and universal increase in LPP amplitudes in this study therefore suggests that reappraisal led participants to engage more attentively with all stimuli, potentially to generate more positive interpretations.

To assess potential confounding due to the block design of the experiment, we again examined condition effects on prestimulus baselines (see Supplemental Materials 2.8). The ERP baseline values in the REAPPRAISE blocks did not systematically differ from those in the VIEW blocks, suggesting that the reported effects are unlikely to reflect baseline artifacts. Moreover, no systematic drift in baseline values was observed either within blocks or across the entire experiment.

Table 3
Experimental Effects on LPP Amplitudes

Model term	β	p
Mean (intercept)	6.33	**<.001**
Negative (vs. neutral)	4.99	**<.001**
Positive (vs. neutral)	4.48	**<.001**
REAPPRAISE (vs. VIEW)	1.51	**.002**
1,500–6,000 ms (vs. 250–1,500 ms)	−4.20	**<.001**
REAPPRAISE × Negative	0.06	.96
REAPPRAISE × Positive	−0.88	.47
Negative × 1,500–6,000 ms	−1.49	.08
Positive × 1,500–6,000 ms	−1.17	.17
REAPPRAISE × 1,500–6,000 ms	−1.08	.12
Negative × REAPPRAISE × 1,500–6,000 ms	1.02	.55
Positive × REAPPRAISE × 1,500–6,000 ms	−0.70	.68
σ^2 (error variance)	293.91	
τ_{00} (random intercept variance)		
Stimulus	4.06	
Participant	13.07	
τ_{11} (random slope variance)		
Participant × Negative	5.03	
Participant × Positive	10.10	
ρ_{01} (random intercept–slope correlation)		
Participant × Negative	0.32	
Participant × Positive	0.21	
ICC (intra-class correlation)	0.06	
N		
Participant	64	
Stimulus	84	
Observations	9,870	
Marginal R^2/conditional R^2	0.03/0.09	

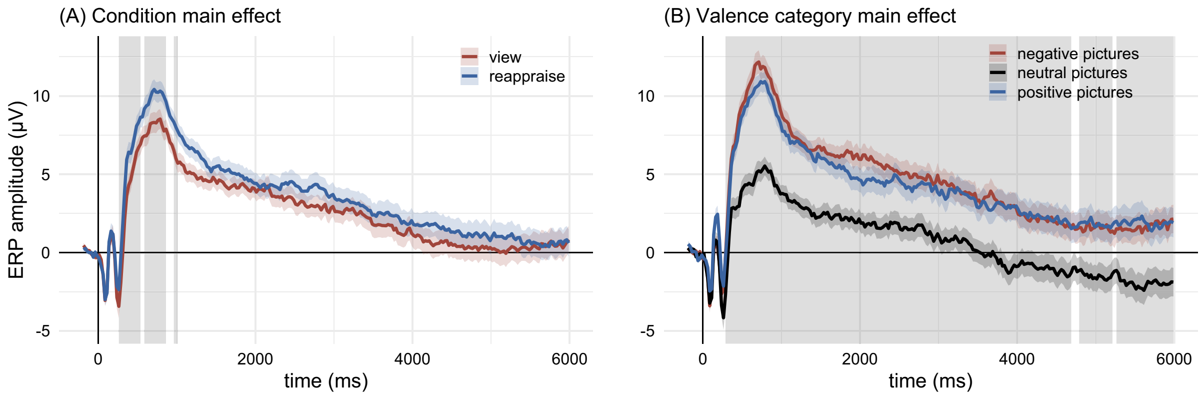
Note. LPP—average amplitude in microvolts between 250–1,500 ms and 1,500–6,000 ms pooled at electrode cites Cz, CP1, CP2, and Pz. Results of the LMM with condition (VIEW, REAPPRAISE), valence category (neutral, negative, positive), and time window (250–1,500, 1,500–6,000 ms) as fixed factors. β = unstandardized beta. All statistically significant p values (<.05) are in bold. LMM = linear mixed model; LPP = late positive potential; CP = central-parietal.

Appraisal Shifts and Affect Ratings

Replicating Study 1, we observed significant shifts in appraisals across all valence categories (see Supplemental Materials 2.7

Figure 6

Time Courses of Condition and Valence Category Effects on LPP Amplitudes



Note. Panel A: The temporal dynamics of ERP amplification for the VIEW (red line) and REAPPRAISE (blue line) conditions. Panel B: The temporal dynamics of ERP amplification for negative (red line), neutral (black line), and positive (blue line) pictures. The ERP waveforms are averaged across Cz, CP1, CP2, and Pz. Gray areas depict significant effects from mass-univariate *t*-tests (panel A) and mass-univariate ANOVAs (panel B), FDR-corrected, $p < .05$, $N = 64$. LPP = late positive potential; CP = central-parietal; FDR = false discovery rate; ERP = event-related potential; ANOVAs = analyses of variance. See the online article for the color version of this figure.

and 2.10), many of which were significantly related to changes in subjective positive and negative affect between the VIEW and REAPPRAISE conditions (see Figure 3 for main and Supplemental Materials 2.9 for additional LCSM coefficients). Based on R^2 values, appraisal shifts explained 57.2%, 25.7%, and 41.8% of the increase in subjective positive affect, as well as 31.2%, 9.5%, and 15.1% of the decrease in subjective negative affect in response to negative, neutral, and positive pictures, respectively.

As in Study 1, we observed the strongest effects for shifts in congruence, which were associated with changes in subjective affect across all valence categories. Effects were most pronounced for negative pictures, where increased congruence was related to higher positive affect ($\beta = 0.78$) and lower negative affect ($\beta = -0.37$). Smaller but similar effects emerged for neutral (positive affect: $\beta = 0.25$; negative affect: $\beta = -0.15$) and positive pictures (negative affect: $\beta = -0.09$). As in Study 1, outlook congruence was associated with changes in subjective affect in some, but not all, valence categories; however, the patterns differed from those observed in Study 1. Specifically, in Study 2, decreased outlook congruence was associated with lower negative affect for neutral pictures ($\beta = 0.14$), whereas increased outlook congruence was associated with higher positive affect for positive pictures ($\beta = 0.33$).

As in Study 1, changes in relevance were again associated with changes in subjective affect, with the pattern varying by valence category; the effects were, however, notably stronger than in Study 1. In Study 2, for negative pictures, decreased relevance was associated with lower negative affect ($\beta = 0.50$). Increased relevance was associated with higher positive affect for both neutral ($\beta = 0.33$) and positive pictures ($\beta = 0.46$).

As in Study 1, only congruence and relevance played a significant role for negative pictures, whereas more diverse patterns emerged for neutral and positive pictures, although the patterns themselves differed slightly from those in Study 1. For neutral pictures, increased controllability was associated with higher positive affect ($\beta = 0.25$). However, unlike in Study 1, changes in current certainty were not related to changes in subjective affect. For positive pictures, decreased other-accountability ($\beta = 0.09$) and increased current certainty ($\beta = -0.19$) were associated with lower negative affect. Unlike in Study 1, controllability shifts were not related to changes in subjective affect for positive pictures. These findings reiterate that reappraisal mechanisms may be context-dependent, engaging a broader range of dimensions in neutral and positive contexts than in negative ones.

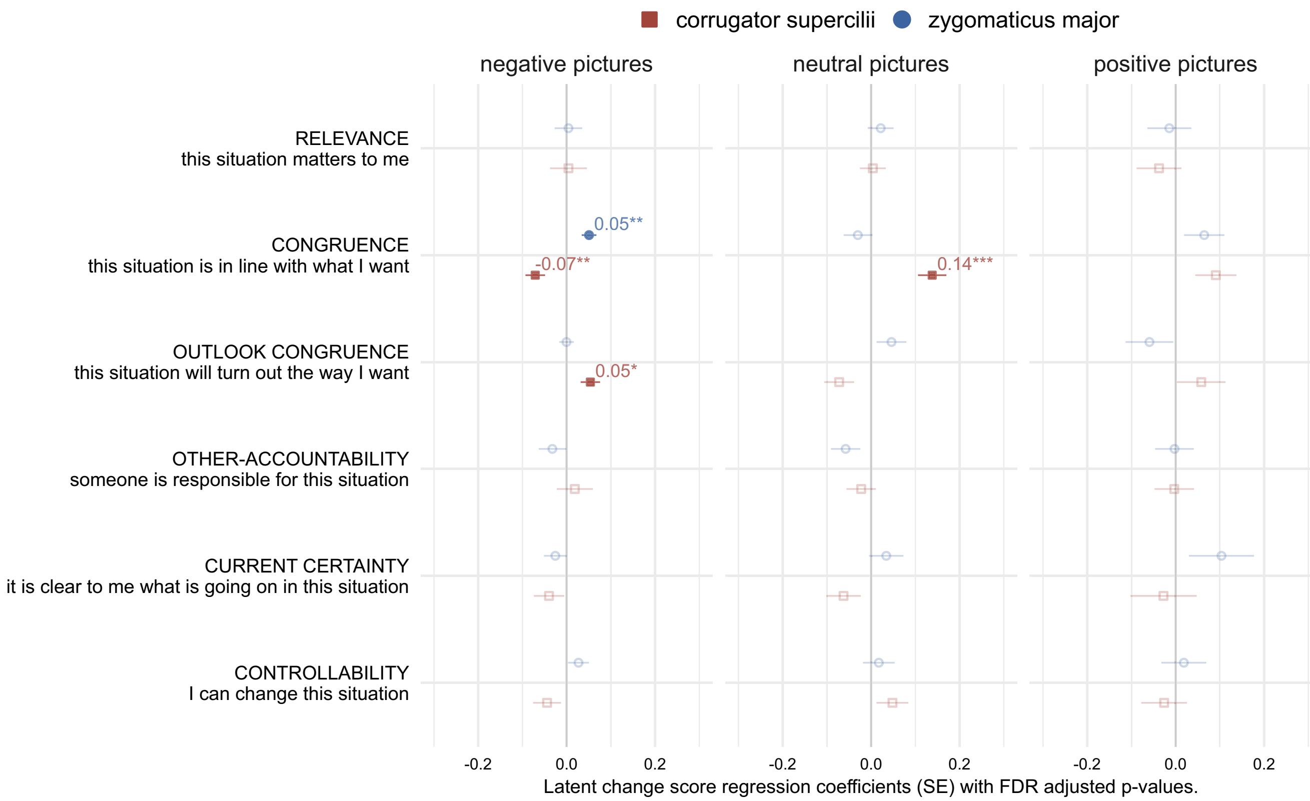
Appraisal Shifts and EMG Reactivity

Several appraisal shifts were related to changes in facial expression between the VIEW and REAPPRAISE conditions (see Figure 7 for main and Supplemental Materials 2.9 for additional LCSM coefficients). However, appraisal shifts explained less variance in changes in facial expression than in changes in subjective affect: 11.7%, 9.3%, and 7.6% of the variance in *zygomaticus* reactivity, and 14.7%, 24.4%, and 4.8% of the variance in *corrugator* reactivity in response to negative, neutral, and positive pictures, respectively. This difference may stem from shared method variance, as both appraisal and subjective affect ratings rely on self-report, which requires cognitive reflection. Experiential, physiological, and behavioral response systems are also shown to have unique sources of variance, limiting convergence across measures (Mauss & Robinson, 2009).

Consistent with appraisal theories (Moors et al., 2013) and subjective affect measures, increased congruence was associated with higher *zygomaticus* reactivity ($\beta = 0.05$) and lower *corrugator* reactivity ($\beta = -0.07$) for negative pictures. However, two unexpected associations emerged: increased outlook congruence for negative pictures and increased congruence for neutral pictures were related to higher *corrugator* reactivity (negative pictures: $\beta = 0.05$; neutral pictures: $\beta = 0.14$). These findings may reflect a link between *corrugator* reactivity and mental effort (Waterink & van Boxtel, 1994), with larger increases in congruence requiring more mental effort and thus leading to higher *corrugator* reactivity despite reduced negative affect.

Figure 7

Associations Between Latent Appraisal Shifts and Latent Changes in Facial Muscle Reactivity



Note. LCSM coefficients representing the associations between a given appraisal shift and changes in zygomaticus major (blue points) and corrugator supercilii (red points) reactivity in response to negative (left column), neutral (middle column), and positive (right column) pictures. Numbers represent unstandardized regression coefficients derived from the LCSMs. Dark filled points indicate statistically significant coefficients. LCSM = latent change score modeling; *SE* = standard error; FDR = false discovery rate. See the online article for the color version of this figure.
Stars denote the levels of FDR-adjusted p values: $^{*} < .05$. $^{**} < .01$. $^{***} < .001$.

The results of the LCSMs for LPP amplitudes are presented in Supplemental Materials 2.9. The associations between latent appraisal shifts and latent changes in LPP are of secondary interest here, as our findings indicate that LPP primarily reflects attentional processes facilitating appraisal shifts rather than affective processes resulting from them.

Overall, the findings of Study 2 replicated the findings of Study 1, suggesting that the efficacy of reappraisal and the involvement of appraisal shifts in reappraisal are largely independent of the limitations and idiosyncrasies of Study 1. Study 2 further extended the findings of Study 1 by demonstrating the efficacy of reappraisal in changing facial expression and highlighting the need for further exploration into the temporal dynamics of facial muscle reactivity, the processing of emotional stimuli during reappraisal using the LPP marker, and the relationship between appraisal shifts and changes in facial expression.

General Discussion

In two complementary studies, we investigated the efficacy of reappraisal aimed at increasing positive affect (i.e., positive goal reappraisal) in response to negative, neutral, and positive stimuli, as well as the mechanistic role of appraisal shifts. In online Study 1, participants rated their subjective positive and negative affect and described how they appraised the depicted situations while viewing and reappraising negative, neutral, and positive pictures. In laboratory Study 2, participants completed the same task, while their facial EMG and EEG signals were recorded as psychophysiological measures of affect.

Efficacy of Reappraisal in Different Valence Contexts

Regarding which valence context is most conducive to reappraisal, we found that positive goal reappraisal improved affect across all valence contexts. Specifically, consistent with prior work (Giuliani et al., 2008; McRae et al., 2012; Samson et al., 2014; Waugh et al., 2016), reappraisal increased subjective positive affect and decreased subjective negative affect in response to negative, neutral, and positive pictures, with the strongest effects observed for negative pictures. In line with previous findings (Kreibig et al., 2023; Li et al., 2018; vanOyen Witvliet et al., 2010), reappraisal also increased *zygomaticus major* reactivity for neutral and positive pictures and decreased *corrugator supercilii* reactivity for negative and neutral pictures, with the strongest effects observed for neutral pictures. The differences we observed between valence categories in reappraisal effectiveness may, in part, reflect floor and ceiling effects. It is therefore reasonable to conclude that positive goal reappraisal can improve affect across all valence contexts, without a systematic advantage in any particular one.

Our findings extend prior research by highlighting the benefits of positive goal reappraisal in neutral contexts, which have largely been overlooked in the literature. Specifically, reappraisal increased both subjective positive affect and *zygomaticus major* reactivity and decreased both subjective negative affect and *corrugator supercilii reactivity* in response to neutral stimuli. As people often experience and aim to improve neutral states (Gasper et al., 2019; Tamir, 2009), reappraisal may offer a valuable tool for generating positive emotions in such contexts and supporting well-being even outside of adversity (McRae & Mauss, 2016). Given that greater reappraisal use is often associated with adaptive outcomes, such as improved psychological well-being (e.g., Gross & John, 2003) and fewer symptoms of psychopathology (e.g., Cludius et al., 2020), our findings underscore its potential to enhance mental health across various valence contexts.

Cognitive Mechanisms of Reappraisal in Different Valence Contexts

Regarding the cognitive mechanisms of reappraisal, we found that appraisal shifts are related to changes in affect during positive goal reappraisal across all valence contexts. A significant proportion of the variance in subjective affect, and a smaller proportion the variance in facial expression, was associated with shifts along various appraisal dimensions in response to negative, neutral, and positive pictures.

Shifts in congruence and relevance occurred across all valence contexts. Increasing congruence during reappraisal led to higher positive affect and lower negative affect across all picture types, consistent with the role of congruence in shaping affective valence (Moors et al., 2013). Increased relevance was associated with higher positive affect for neutral and positive pictures, while decreased relevance was related to lower negative affect for negative and neutral pictures. This aligns with the proposed role of relevance in shaping affective intensity (Moors et al., 2013; Yeo & Ong, 2024). Overall, different combinations of shifts in congruence and relevance may serve both qualitative (i.e., changing emotion type) and quantitative (i.e., changing emotion intensity) emotion regulation goals (Kreibig et al., 2023). Specifically, changing congruence may support both types of regulation goals, whereas changing relevance may serve primarily quantitative goals.

We also observed differences in appraisal shifts across valence contexts. Specifically, while only congruence and relevance played a role in negative pictures, additional dimensions emerged as significant for neutral and positive pictures. A broader range of shifts may be beneficial in neutral and positive contexts, possibly because these contexts elicit more idiosyncratic appraisals, allowing for more nuanced reappraisals. This suggests that reappraisal mechanisms may vary depending on the valence context. Given the importance of flexibility in adaptive emotion regulation (e.g., Aldao & Tull, 2015; Petrova & Gross, 2023), our findings highlight the value of adjusting appraisal shifts to suit different contexts.

Our findings complement prior evidence on the role of appraisal shifts in reappraisal. Whereas existing studies have documented appraisal shifts involved in reducing negative affect elicited by mostly negative life events (A. Uusberg et al., 2023), the present findings extend this research to positive and neutral contexts. Moreover, in the prior studies, participants appraised and then reappraised the same stimuli, potentially conflating reappraisal effects with affective adaptation and demand effects. This limitation was minimized in the present study, as participants engaged with equivalent but different stimuli in the VIEW and REAPPRAISE conditions.

This line of research is beginning to substantiate the proposal that appraisal shifts play a mechanistic role in translating the cognitive changes involved in reappraisal into changes in affect (A. Uusberg et al., 2019). Measuring appraisal shifts could therefore provide a systematic way to characterize different forms of reappraisal, addressing the limited overlap among existing taxonomies (e.g., Garnefski & Kraaij, 2007; McRae et al., 2012; Vishkin et al., 2020). Describing reappraisal in terms of appraisal shifts may also facilitate the development of interventions that promote reappraisal use. Our findings suggest that, in general, interventions targeting congruence and relevance may result in strongest and most universal effects. However, we also found context-dependent shifts, suggesting that reappraisal interventions might benefit from being customized by tailoring recommended appraisal shifts to individual characteristics and contextual demands (Aldao & Tull, 2015; Troy et al., 2013).

Limitations and Future Directions

Our investigation of positive goal reappraisal in response to negative, neutral, and positive stimuli provided valuable insights into its efficacy and underlying cognitive mechanisms. Nonetheless, several limitations need to be addressed, which highlight important directions for future research.

Our studies focused on the five most common appraisal dimensions, which can be considered necessary but not sufficient to capture the full range and nuances of appraisals influencing affect (Moors et al., 2013; Yeo & Ong, 2024). Future research could expand the scope by including additional dimensions—for example, by distinguishing intrinsic pleasantness from congruence or urgency from relevance. Additionally, future work could investigate idiosyncratic response patterns in appraisal shifts by measuring appraisals both before and after reappraisal to obtain trial-by-trial approximations of these shifts. Such data could also help explore the relationship between appraisal shifts and psychophysiological measures of affect. Furthermore, understanding how different appraisal dimensions function independently is crucial for developing targeted interventions, making experimental studies focusing on specific dimensions an important direction for future research.

Another limitation is that our focus was exclusively on reappraisal aimed at increasing positive affect, an understudied goal compared to the more commonly examined goal of decreasing negative affect. However, adaptive emotion regulation requires flexibility in both decreasing and increasing negative and positive affect (Aldao, 2013; Aldao et al., 2015). Existing research indicates, for instance, that qualitative goals, which aim to increase the impact of the opposite valence, are more effective in modulating both negative and positive affect compared to quantitative goals, which aim to decrease the predominant affective impact (Kreibig et al., 2023). Although it fell beyond the scope of our research, future studies should explore the effects and mechanisms of different emotion regulation goals across various valence contexts, rather than focusing solely on increasing positive affect.

We also identified findings that merit further investigation. First, in contrast to previous research (e.g., Kreibig et al., 2023; vanOyen Witvliet et al., 2010), we did not observe increased *zygomaticus* reactivity for negative pictures. This may be because a qualitative change in affect from negative to positive requires more time than the 6-s window used in this study. Second, we found that positive goal reappraisal increased early LPP amplitudes, regardless of valence category. The overlap of this early LPP amplification with the ERP component P300, along with the absence of an interaction between condition and valence category, suggests that the effect may reflect attentional processes rather than changes in arousal alone. Together, these findings highlight the need for further research on how positive goal reappraisal impacts the processing of emotional stimuli.

Constraints on Generality

The sample in Study 1 was diverse in gender, age, and geographic location. While ethnically heterogeneous, individuals identifying as White were overrepresented. In contrast, Study 2 featured a relatively homogeneous sample of White university students. As the findings of Study 2 largely replicated those of Study 1, the results demonstrate a fair degree of generalizability. However, the observed patterns may not extend to individuals from different sociodemographic and cultural backgrounds. Enhancing the generalizability of these findings would require larger and more diverse samples.

The generalizability of our findings is also limited by the use of affective pictures from standardized databases to elicit affect in both studies. While this approach is common in experimental emotion regulation research, generalizability could be improved by using a broader range of stimuli (e.g., film clips or autobiographical memories) and incorporating diverse methods (e.g., ecological momentary assessments). Moreover, the block design further constrains generalizability, as it may induce a response set in which individuals consistently apply one regulation strategy and anticipate its use across trials, complicating the interpretation of both subjective and psychophysiological measures. Although LPP baselines did not differ between the VIEW and REAPPRAISE blocks, EMG baselines did, in the same direction as the reappraisal effects on EMG reactivity. This suggests that studies using trial-by-trial design might observe different, potentially larger, reappraisal effects. While the block design was chosen to reduce the cognitive load of repeated appraisal self-reports, a trial-by-trial design in future research could better capture strategy-related effects and test the robustness of the findings under greater variability and flexibility.

Additionally, the validity of the self-report measures of appraisals used here is limited, as appraisals can occur automatically and may not always be accessible to conscious awareness (e.g., Scherer, 1999; Smith & Lazarus, 1993). Although direct measurement of appraisals is challenging, generalizability could be enhanced by complementing self-reports with indirect measures, such as inferring appraisals from verbal expressions using natural language processing algorithms (e.g., Troiano et al., 2023).

While several constraints on generality should be addressed in future research, the current findings remain encouraging. Both studies demonstrated substantial overlap in results, despite using samples from different linguistic contexts and distinct measures of affect. This convergence across methodological differences strengthens confidence in the observed effects and suggests that the findings may extend beyond the specific contexts examined here.

Conclusions

In two complementary studies, we found that positive goal reappraisal increased subjective positive affect and *zygomaticus major* reactivity, while decreasing subjective negative affect and *corrugator supercilii* reactivity. These findings suggest that positive goal reappraisal may offer benefits across all valence categories. Furthermore, appraisal shifts, particularly changes in congruence and relevance, were associated with reappraisal-related changes in different affect measures. Reappraisal also engaged a broader range of dimensions in neutral and positive contexts than in negative ones, suggesting that reappraisal mechanisms may be context-dependent. These findings contribute to the growing literature on reappraisal and will hopefully encourage other researchers to further explore the mechanistic role of appraisal shifts in reappraisal.

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